ADDIS ABABA UNIVERSITY
DEPARTMENT OF MATHEMATICS
Mathematics for Natural Sciences (Math 1011)
Worksheet III and IV, December 2025

Chapter 3: FUNCTIONS

(1) Let $A = \{1, 2, 3, 4\}$ and R be a relation on A such that $\{(x, y)|y - x^2 + 2x \leq 0\}$. Describe R by complete listing method and find its domain and range.

(2) In each case determine whether f is well-defined. Give reason for your answer
(a) $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = -n$, $\forall n \in \mathbb{Z}$ (b)$f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sqrt{x}$, $\forall x \in \mathbb{R}$
(c) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \frac{1}{x}$, $\forall x \in \mathbb{R}$ (d) $f : \mathbb{Q} \to \mathbb{Z}$, $f(\frac{m}{n}) = mn$, $\forall m, n \in \mathbb{Z}, n \neq 0$

(3) Which of the following relations are functions? Which relations have their inverse that are functions?
(a) $R_1 = \{(1, 2), (2, 3), (3, 1), (4, 3)\}$. (b) $R_2 = \{(x, y)|x, y \in \mathbb{Z} \ \wedge \ x - y \text{ is even}\}$.
(c) $R_3 = \{(x, y)|x, y \in \mathbb{R} \ \wedge \ x - y = 4\}$.

(4) Let $A = \{a, b, c, d\}$, $B = \{a, c, d, e\}$. Let $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$ such that $f = \{(a, 1), (b, 2), (c, 2), (d, 4)\}$ and $g = \{(a, 0), (c, 4), (d, 8), (e, 0)\}$. State the domain and find:
(a) $f + g$ (b) $f - g$ (c) $f.g$ (d) f/g.

(5) (i) Given $f(x) = x^2 + 1$ and $g(x) = \sqrt{x - 1}$, state the domain and find
(a) $f \circ g$ (b) $g \circ f$ (c) $f \circ f$ (d) $f \circ g \circ f$
(ii) Given a set A and $f : A \to A$, $g : A \to A$. Is $f \circ g = g \circ f$?

(6) Which of the following functions are one-to-one, onto and one-to-one correspondence?
(a) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^n$, n is odd (b) $k : [0, \infty) \to [1, \infty)$, $k(x) = x^2 + 1$.
(c) $h : E^+ \to \mathbb{N}$, $h(n) = \frac{3n+2}{2}$ where $E^+ = \{2n|n \in \mathbb{N}\}$ (d) $g : \mathbb{Z} \to \mathbb{N}$, $g(n) = |n|$

(7) Determine whether each of the following functions has an inverse or not. If an inverse exists, give the domain and the range of the inverse.
(a) $f(x) = x^5 + 1$ (b) $f(x) = x + |x|$ (c) $f(x) = x + \sin x$ (d) $f(x) = \ln(2 - x)$

(8) Find all intercepts, asymptotes and sketch the graph of the following rational functions. Find the domain and range of each.
(a) $f(x) = \frac{x}{x^2-4}$ (b) $f(x) = \frac{x^2+1}{x^2-1}$ (c) $f(x) = \frac{x^2-x}{x-3}$ (d) $f(x) = \frac{x+1}{x^2-1}$

(9) Find the domain of the following functions
(a) $f(x) = \ln(x^2 - 2x - 15)$ (b) $g(x) = \frac{x}{\ln(5x+4)}$

(10) Find the inverse of the following functions
(a) $y = \ln(3x + 7)$ (b) $y = 2^{x+3} - 1$
(c) $y = (4^x - 1)^{\frac{1}{4}}$ (d) $y = \log_4(3^x + 4)$

(11) Solve the following equations
(a) $5^{2x} - 2(5^x) - 15 = 0$ (b) $\ln(n^2 + 12) = \ln(-9n - 2)$
(c) $\log_9(-11x + 2) = \log_9(x^2 + 30)$ (d) $\log_5 6 + \log_5 2x^2 = \log_5 48$

(12) $15,000$ Birr is invested in an account that yields 5% interest per year. After how many years will the account be worth $91,221.04$ Birr if the interest is compounded yearly?

(13) A function f is said to be even if $f(-x) = f(x)$ for all x in the domain of f and f is said to be odd if $f(-x) = -f(x)$ for all x in the domain of f. Classify the following functions as even, odd and neither.
(a) $f(x) = x \cos x - \tan x$ (b) $f(x) = \cot^2 x - \sin x$
(c) $f(x) = \sec x \csc x$ (d) $f(x) = x \sinh x - \cosh x$

(14) If α and β are angles of the first quadrant such that $\tan \alpha = \frac{3}{4}$ and $\tan \beta = \frac{1}{7}$, find the quadrant in which $\alpha + \beta, \alpha - \beta, 2\alpha$ and 3β terminate and calculate
(a) $\sin(\alpha + \beta)$ (b) $\cos(\alpha - \beta)$
(c) $\csc(2\beta)$ (d) $\sec(\frac{\alpha}{2})$

(15) Find the period and amplitude, and sketch the graph of the following equations
(a) $y = 3 \sec x$ (b) $y = \sin 4(x - \frac{\pi}{2})$
(c) $y = \cos \frac{1}{3}x$ (d) $y = \tan(x - \frac{\pi}{4})$

(16) Prove whether the following are identities
  (a) $\sin 3x = 4\cos^3 x - 3\cos x$ (b) $\frac{\cos x}{1-\sin x} = \frac{1+\sin x}{\cos x}$
(17) Solve the following equations in $[0, 2\pi]$
  (a) $\cot^2\theta \sin\theta - 3\sin\theta + \cos^2\theta - 3 = 0$ (b) $2\cos^4\theta + 9\sin^2\theta - 5 = 0$
  (c) $\sin\theta + 2\cos\theta = 1$ (d) $2\cos\theta + \sqrt{4 + \csc^2\theta} = 0$
(18) Sketch the graph of the following functions
  (a) $y = 3\cos 2x$ for $0 \le x \le 2\pi$ (b)$y = 3\sin 2(x + \frac{\pi}{4})$ for $-\frac{\pi}{4} \le x \le 2\pi$
(19) Prove whether the following are identities
  (a) $\frac{2\tanh\frac{1}{2}x}{1-\tanh^2\frac{1}{2}x} = \sinh x$ (b) $\frac{\cosh(\ln x)+\sinh(\ln x)}{\cosh(\ln x)-\sinh(\ln x)} = x^2$

Chapter 4: ANALYTIC GEOMETRY

(1) Find the distance between the point $A(2, -3)$ and the line with equation $L : 3x - 4y + 2 = 0$.
(2) The point P divides the line segment AB joining the points $A(2, 1)$ and $B(-3, 6)$ in the ratio $2 : 3$. Does the point P lie on the line with equation $x - 5y + 15 = 0$? Justify
(3) What is the equation of the locus of a point which moves such that 4 times its distance from the x-axis is the square of its distance from the origin?
(4) Find the equation of the circle if two of it's diameters are $x + 4y = 5$ and $7x - y = 6$, and whose radius is $\sqrt{7}$.
(5) Find the equation of the circle which passes through the points $(1, 2)$, $(2, 2)$ and $(4, -1)$.
(6) Find the equation of the circle
  (a) with diameter having end-points $(-8, 5)$ and $(8, -1)$
  (b) with center $(-4, 1)$ and tangent to the line $3x + 4y = 10$
  (c) passing through the points $(-1, 1)$, $(3, 2)$ and whose center lies on the line $x - 2y + 2 = 0$
(7) Find the equation of the parabola
  (a) with vertex at $(0, 0)$ and focus $(0, -6)$. Sketch the curve
  (b) with focus at $(5, 0)$ and equation of the directrix $x = -5$
  (c) with end points of its latus rectum at $(16, 8)$ and $(-16, 8)$ (two possible solutions)
  (d) symmetric about y-axis, vertex at origin and passes through $(6, -8)$
(8) Find the vertex, focus, the axis, directrix, latus rectum and length of latus rectum of the following parabola.
  (a) $y^2 = 12x$ (b) $3(x - 2)^2 + 2y = 0$
(9) Find the equation of the the ellipse
  (a) with center at $(-2, 1)$, a focus at $(-2, 4)$ and a vertex at $(1, 1)$
  (b) with foci at $(2, -2)$ and $(4, -2)$
  (c) with vertices at $(1, -2)$ and $(5, -2)$ (two possibilities)
(10) Find the foci, the transverse axis, the conjugate axis, the vertices, the latus rectum and the eccentricity of the following ellipse.
  (a) $25x^2 + 4y^2 = 100$ (b) $(\frac{x}{2} - 1)^2 + (\frac{y}{3} - 1)^2 = 1$
(11) Find the equation of the the hyperbola
  (a) with vertices at $(9, -3)$ and $(-5, -3)$ and foci at $(\sqrt{53}, -3)$ and $(-\sqrt{53}, -3)$
  (b) with vertices at $(-5, 0)$ and $(5, 0)$ and conjugate axis of length 12
(12) Find the foci, the major axis, the minor axis, the vertices, the latus rectum and the eccentricity of the following hyperbola.
  (a) $4x^2 - 25y^2 = -1$ (b) $(y - 3)^2 - \frac{(x+2)^2}{9} = 1$
(13) Assuming that the following conic sections are not degenerate, classify the conic sections whose Cartesian equations are
  (a) $y^2 - 4y + 16 = 4x^2 + 8x$ (b) $2x^2 - 5xy + 3y^2 - x + y = 4$
  (c) $4y^2 - 4y + 28x + 29 = 0$ (d) $4x^2 + y^2 - 6x + 5 = 0$
(14) A satellite dish is shaped like a paraboloid. The signals that emanate from a satellite strike the surface of the dish and are reflected to a certain point where the receiver is located. If the dish is 12 feet across its opening, and is 4.5 feet deep at its center, how far from the center of the dish should the receiver be placed?